A Framework for the Mathematical Analysis of AI Papers based on maths primitives

By

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Background

Important context

This is a framework for analysing the mathematical contribution of AI papers. AI papers have multiple dimensions This framework evaluates only one dimension of an AI paper: its mathematical contribution. Many highly influential AI papers derive their novelty from multiple dimensions such as architecture, systems engineering, datasets, training methodology, evaluation, or product design rather than mathematics. These require complementary analytical frameworks / primitives. However, the mathematics is a common framework / starting point even when the emphasis of the paper lies beyond the mathematics . The framework is designed tobe used with LLM tools.

It is important to emphasise that this is an educational / learning framework. The primary audience is students or developers who are typically not in academia but have an interest in understanding AI from a mathematical standpoint through AI papers.

Introduction

The aim of this paper is to create a framework for the analysis of an AI paper from a mathematical perspective. The initiative is related to the course [Mathematical foundations of Artificial Intelligence]

The paper is designed with an educational focus - enabling the practitioner to create an initial detailed analysis of a paper from a mathematical perspective through a set of questions that can be asked.

AI papers vary enormously in their mathematical character. The framework below proceeds through six layers by identifying a set of mathematical primitives for analysing AI papers These include : object identification, theoretical frame, claim structure, proof architecture, the empirical–theoretical bridge, and a methodological audit.

Each layer asks a distinct kind of question, and each has a characteristic set of primitives — the possible objects, frames, claim-types, proof techniques, gap-patterns, and failure modes that the analyst learns to recognise. Identifying which primitives are operative at each layer is the operational core of the framework.

To understand the context, we first understand the landscape of AI papers.

Understanding the landscape of AI papers

We can divide AI papers into several dimensions

Dimension 1 — Pure mathematical AI papers

The framework addresses this category. Examples include: Generalisation theory, Statistical learning theory, PAC-Bayes, VC theory, Approximation theory, NTK papers, Optimisation theory, Information bottleneck Scaling law theory etc.

Dimension 2 — Modern foundation model theory

Examples of papers in this category include transformers, attention, in-context learning etc. These often mix optimisation, probability, geometry, information theory, empirical validation. The mathematical framework also has synergies here

Dimension 3 — Reinforcement Learning

For Reinforcement learning typical mathematical objects become policies, value functions, Bellman operators, Markov decision processes, games. Layer 2 in the framework naturally maps onto optimisation, probability, game theory, dynamical systems

Dimension 4 — Bayesian AI

Many Bayesian ideas align based on

- posterior distributions
- variational inference
- uncertainty
- priors
- approximate inference

Much of Layer 4 is already Bayesian mathematics.

Dimension 5 — Mechanistic interpretability

Traditional explainability asks Why did the model produce this prediction?

Mechanistic interpretability asks something much deeper: **What algorithm has the neural network actually learned?**

Instead of treating GPT as a black box, it attempts to reverse engineer the computation.

Think of it as **Computer architecture meets neuroscience meets software reverse engineering.**

Consider an analogy

Imagine someone gives you a compiled executable. Traditional explainability asks Why did the program output 42? Mechanistic interpretability asks What functions exist inside the binary? What data structures? What algorithms? What modules?

Typical mathematical objects considered are neurons, features, directions, subspaces, attention heads, circuits, activation vectors, sparse features, residual stream geometry, activation manifolds

These are computational objects rather than statistical objects. Example papers

A Mathematical Framework for Transformer Circuits

Interpretability in the Wild

Sparse Autoencoders (SAEs)

Interestingly, Mechanistic Interpretability uses very different mathematics from statistical learning theory.

Instead you see

- Linear algebra for Activation vectors
- Geometry Feature for manifolds
- Graph theory for Circuits
- Information theory for Representations
- Sparse optimisation for Sparse autoencoders
- Matrix factorisation for Features
- Tensor decomposition for Hidden representations
- Causal analysis for Activation patching

Dimension 6 — Architecture papers

These papers typically suggest new architectures

Examples include the mamba paper, mixture of experts paper . Here, we have other considerations such as implementation, GPU efficiency, kernels, engineering trade-offs

Dimension 7 — Systems papers

This category is actually one of the fastest-growing areas in AI research. These papers are typically published atsystems conferences such as **MLSys**, **OSDI**, **NSDI**, **ASPLOS**, **ISCA**, and **SC**, rather than at purely machine learning conferences.

1. Distributed Training

These papers study **how to train trillion-parameter models efficiently**. The mathematics is usually straightforward. The novelty lies in communication, GPU utilisation, memory, parallelism, scheduling

Examples include:

Horovod: Fast and Easy Distributed Deep Learning in TensorFlow

ZeRO: Memory Optimizations Toward Training Trillion Parameter Models

Megatron-LM

GSPMD: General and Scalable Parallelization for ML Computation Graphs

Pathways: Asynchronous Distributed Dataflow for ML - Google

2. Serving Infrastructure

These papers ask **How do we serve an LLM to millions of users?**

Training is finished. Deployment begins. Typical questions these papers address include latency,batching, scheduling, cache management, memory management

Orca: A Distributed Serving System for Transformer-Based Generative Models

vLLM Efficient Memory Management for Large Language Model Serving with PagedAttention

TensorRT-LLM

JellyBean Serving ML workflows efficiently across cloud and edge infrastructure.

3. Compiler Optimisation

These papers ask **How can compilers automatically generate faster AI programs?**

Instead of inventing new neural networks, they invent better compilers.

Ex

MLIR: A Compiler Infrastructure for the End of Moore's Law

XLA compiler paper

TVM: An Automated End-to-End Optimizing Compiler for Deep Learning

Triton from OpenAI - GPU programming language

CompilerGym

4. Inference Engines

These papers ask

How can we make trained models run faster?

Typical goals include lower latency, higher throughput, less memory, cheaper inference

FasterTransformer

Exploring TensorRT to Improve Real-Time Inference for Deep Learning

ONNX Runtime

ggml-org/llama.cpp: LLM inference in C/C++

In essence, A systems paper is asking a completely different question i.e. Can we make it run efficiently at scale?. The DeepSeek papers are a perfect example. The DeepSeek technical reports contain some mathematical analysis, but much of their impact comes from systems innovation.

Dimension 8 — Dataset papers

Examples ImageNet, Improved methodology for longitudinal Web analytics using Common Crawl , LAION LAION-5B: An open large-scale dataset for training next generation image-text models

The mathematics is minimal. The contribution is collection, curation, statistics, annotation, diversity

Dimension 9 — Benchmark papers

Examples MMLU, SWE-bench, . Again, the contribution is evaluation, methodology, metrics rather than mathematical theory.

Dimension 10 — AI Governance / Policy

These papers are about institutions, economics , regulation, ethics - rather than mathematics. So, the framework is not applicable here. It would not pretend to evaluate distributed systems. That is actually a strength.

Understanding Novelty in AI papers

Every AI paper has a primary object of novelty and could be represented along multiple novelty dimensions. For example, based on our above discussion, novelty dimensions could be :

- Mathematical
- Algorithmic
- Architectural
- Systems
- Data
- Training Methodology
- Evaluation
- Product/Deployment

Each of these dimensions could be represented by primitives

Mathematical Framework primitives

- Mathematical object
- Theoretical frame
- Claims
- Proof architecture
- Theory ↔ Practice
- Audit

Systems Framework primitives

- Parallelisation strategy
- Memory architecture
- Communication
- Scheduling
- Fault tolerance
- GPU utilisation
- Scaling efficiency
- Cost

Architecture Framework primitives

- Computational graph
- Inductive bias
- Information flow
- Attention mechanisms
- Memory
- Compositionality
- Expressivity
- Scaling

Training Framework primitives

- Data mixture
- Curriculum
- Optimiser
- RL
- Alignment
- Distillation
- Hyperparameters
- Compute allocation

etc

Novelty is multidimensional

The AI community often asks: "Is this paper novel?" But that's actually the wrong question. The better question is: **"Novel in what sense?"** That immediately removes a huge amount of confusion. DeepSeek is novel. But not for the same reasons that PAC Learning is novel.

How to Read Artificial Intelligence Research

Therefore, in a complete sense, to read an AI paper, we should analyse it from the perspective of multiple dimensions and explore the novelty or primarily intellectual contribution within the dimension.

With this background, we can now consider a

Overview of six layer primitives for Mathematical analysis of AI papers

We first explain these six layers and then expand on each layer in more detail

Layer 1: Identifying the Mathematical Object

The first layer asks what mathematical object the paper is in fact studying. AI papers frequently obscure this, talking about models, systems, or agents without specifying the underlying mathematical structure. The first analytical move is to recover that structure.

The primitives at this layer are the recognised kinds of mathematical objects that appear as the central entity in AI papers.

Empirical risk minimiser over a hypothesis class. The classical statistical-learning object: a function class, a loss, a sample from an unknown distribution, and the minimiser of empirical risk within the class. The object underlying VC theory, Rademacher complexity, and PAC-Bayes analyses.

The trained network as a specific function. A particular function in some functional space, characterised by analytic properties such as Lipschitz constant, spectral norms of weight matrices, and margin distributions. The object underlying margin-based generalisation bounds and norm-based capacity measures.

The parameter trajectory during training. A curve through parameter space, traced out by gradient descent or its stochastic variants. The object of dynamical-systems analyses

— edge-of-stability phenomena, saddle-point escape, mode connectivity, the linear interpolation between trained networks.

The distribution over parameters. A probability measure on parameter space, evolved by the training procedure. The object of mean-field analyses, treating the ensemble of neurons or particles as the relevant entity. Analysis proceeds via partial differential equations on the space of measures.

Network-induced kernels. Kernels constructed from the network — the Neural Tangent Kernel (inner products of parameter-gradients), the conjugate kernel (inner products of pre-activations at the final layer), the path kernel. The object that brings reproducing-kernel-Hilbert-space machinery to bear on neural networks.

Information-theoretic quantities of internal representations. Mutual information between layers, channel capacities of intermediate representations, rate–distortion functions for hidden activations. The object of the information-bottleneck literature.

The loss landscape as a geometric object. The graph of the loss function over parameter space, studied for its critical points, their Hessians, the connectivity of low-loss regions, and the topology of level sets.

The data distribution and its transformations. The probability distribution generating data, viewed via a generative process. In diffusion models, a stochastic differential equation; in normalising flows, an invertible map; in score-based models, a vector field; in autoregressive models, a conditional factorisation.

The optimiser as a dynamical or stochastic object. SGD itself as the entity studied — its stationary distribution, its mixing time, its escape behaviour from local minima, its implicit regularisation. The object of analyses that focus on the algorithm rather than its output.

Mechanistic computational objects. Circuits, features, attention heads, induction patterns, residual-stream subspaces. The entities the mechanistic interpretability literature has been working to define precisely enough to serve as mathematical objects rather than informal labels.

Scaling laws as parametric families. The loss as a power-law function of compute, parameters, and data. An empirically-derived parametric family rather than an object derived from a specified hypothesis class — mathematically tractable in its own way.

Game-theoretic equilibria. Nash, correlated, or Stackelberg equilibria between learners and adversaries, generators and discriminators, players in multi-agent settings.

Bayesian posteriors. Distributions over hypotheses given data, with associated posterior predictives, marginal likelihoods, and information-theoretic quantities. The object of

Bayesian neural networks, Bayesian model selection, and the wider probabilistic-machine-learning tradition.

The analytical move at this layer is to determine which of these objects (or which combination) the paper is in fact studying. The hidden trap is that many AI papers operate in a setting where this question is left implicit — the data distribution is unstated, the loss is a proxy for what the authors actually care about, the hypothesis class is whatever a particular architecture produces under SGD. The analyst records what is specified and what is implicit, because the implicit parts are usually where the theoretical action lies.

Layer 2: Theoretical Frame

The second layer asks which mathematical tradition the paper is operating within. The object having been identified, the choice of frame determines what kinds of questions can be asked about it and what kinds of result can be obtained.

The primitives at this layer are the recognised theoretical traditions that AI papers draw on.

Statistical learning theory. The frame organised around generalisation: PAC learning, VC theory, Rademacher complexity, PAC-Bayes, algorithmic stability. Asks how training error relates to test error.

Optimisation theory. The frame organised around convergence: convex and non-convex analysis, stochastic approximation, variance reduction, accelerated methods, variational analysis. Asks whether and how a procedure reaches a target.

Information theory. The frame organised around uncertainty and channels: entropy, mutual information, divergences, rate–distortion, source coding, channel capacity. Asks what can be transmitted, compressed, or distinguished.

Approximation theory. The frame organised around representability: universal approximation, depth–width trade-offs, expressivity, approximation rates. Asks what can in principle be represented by a function class.

Dynamical systems. The frame organised around evolution: continuous and discrete dynamical systems, attractors, bifurcations, Lyapunov stability. Asks how a system evolves and where it tends.

Geometry and topology. The frame organised around shape and structure: differential geometry, Riemannian manifolds, algebraic topology, persistent homology. Asks about curvature, connectivity, and invariants of spaces.

Probability theory. The frame organised around random structure: concentration, large deviations, martingales, mean-field analysis, propagation of chaos. Asks how randomness aggregates and concentrates.

Computational complexity. The frame organised around resources: time complexity, space complexity, query complexity, communication complexity, reductions. Asks what can be computed and at what cost.

Statistical physics. The frame organised around emergent collective behaviour: phase transitions, free energy, replica method, spin glasses. Asks how macroscopic regularities arise from microscopic interactions.

Bayesian inference. The frame organised around prior–posterior reasoning: priors, likelihoods, posteriors, marginal likelihoods, Bayesian model averaging. Asks how beliefs should be updated by evidence.

Causal inference. The frame organised around interventions and counterfactuals: structural causal models, do-calculus, identifiability of causal effects. Asks what would happen under intervention.

Game theory. The frame organised around strategic interaction: Nash equilibria, regret, no-regret learning, mechanism design. Asks what occurs when multiple optimising agents interact.

The analytical move at this layer is to locate the paper within these frames and to note which frames are conspicuously absent. A paper offering an optimisation result alone, while implying generalisation, is sitting in the gap between two frames — and the gap is where most of the analytical action lies. The frame also dictates what counts as a meaningful claim: a bound that is impressive in approximation theory may be vacuous in statistical learning theory; an optimisation result may be unmoored from any generalisation implication.

Layer 3: Claim Structure

The third layer classifies the paper's claims along four axes: type, strength, conditions, and scope. The primitives at this layer are organised by axis.

Types of claim: The recognised kinds of mathematical claim that AI papers make.
Existence. There exists a function, configuration, or solution achieving X — typically non-constructive.

Convergence. Some procedure tends to a target as iterations or data grow.

Generalisation. Test-time performance is bounded in terms of training-time quantities.

Sample complexity. The number of samples required to achieve a target performance.

Computational complexity. The time, space, or query cost of an algorithm or problem.

Expressivity. The class of functions that can be represented within a model family.

Identifiability. Parameters or latent structures can be recovered uniquely (or up to a stated equivalence) from data.

Stability and robustness. Small perturbations to inputs, parameters, or data produce bounded changes in outputs.

Approximation rate. How the approximation error decays as a resource (depth, width, samples) grows.

Lower bound or impossibility. No procedure of a specified kind can achieve better than a stated performance.

Equivalence or reduction. Two settings are mathematically equivalent, or one reduces to the other.

Calibration. Predicted probabilities track empirical frequencies in a quantifiable sense.

Strength dimensions: Each claim is positioned along several dimensions of strength: worst-case versus average-case; asymptotic versus finite-sample; distribution-free versus distribution-dependent; high-probability versus in-expectation; tight versus loose. A claim's strength is its joint position along these dimensions; a "high-probability finite-sample distribution-dependent tight" bound is a different kind of claim from an "in-expectation asymptotic distribution-free loose" one, even if both are framed as the same theorem type.

Categories of condition: The hypotheses on which claims rest, organised by kind.

Regularity conditions. Continuity, smoothness, Lipschitzness, boundedness, differentiability.

Statistical conditions. I.i.d. sampling, bounded moments, sub-Gaussian or sub-exponential tails, mixing.

Structural conditions. Sparsity, low rank, manifold hypothesis, separability, low intrinsic dimension.

Architectural conditions. Width, depth, activation function, parameterisation, initialisation scheme.

Algorithmic conditions. Learning rate, batch size, regularisation, number of training steps, optimiser choice.

Scope-gap types: The gap between what the theorem formally proves and what the paper claims discursively.

Algorithmic idealisation. Gradient flow rather than SGD; exact arithmetic rather than finite precision; full-batch rather than mini-batch.

Architectural idealisation. Shallow rather than deep; linear rather than nonlinear; specific simplified family rather than general architecture.

Limit-taking. Infinite width, infinite data, infinite time, vanishing learning rate.

Distributional simplification. Gaussian rather than realistic; isotropic rather than structured; bounded support; specific generative process.

The analytical move at this layer is to record, for each claim in the paper, where it sits along each axis. The honest reading flags exactly where each condition is doing work in the proof and exactly where the gap between idealisation and practice lies.

Layer 4: Proof Architecture

The fourth layer examines the structure of the proof — not by verifying it line by line, but by identifying the load-bearing argument and the standard machinery deployed.

The primitives at this layer are the recognised proof techniques used in AI mathematics.

Concentration inequalities. Hoeffding, McDiarmid, Bernstein, Bennett, Talagrand. The workhorse of generalisation theory: bounded random quantities concentrate around their expectations at controllable rates.

Covering and chaining. Discretising a function class via *\varepsilon*-nets and using the discretisation to control supremum quantities. The basis of Dudley-style entropy bounds and Rademacher complexity proofs.

Symmetrisation. Replacing dependence on the data distribution with dependence on independent copies, then handling the symmetric structure. Standard in Rademacher complexity proofs.

Coupling arguments. Constructing a joint distribution of two random objects to control their distance. Used in Markov chain mixing, in stability arguments, in convergence of stochastic processes.

Martingale arguments. Treating a sequence as a martingale and applying concentration or convergence results. Central to online learning regret analyses.

Fixed-point theorems. Banach, Brouwer, Schauder, Kakutani. Used in equilibrium analyses, in convergence to stationary distributions, in GAN convergence arguments.

Variational principles. Recasting a problem as an extremisation over a function or distribution. The basis of variational inference, evidence lower bounds, and duality arguments in optimisation.

Lyapunov function arguments. Constructing a function that decreases along the dynamics, witnessing convergence or stability. Used in training-dynamics analyses.

Markov chain analyses. Mixing times, spectral gaps, stationary distributions. Used in SGD-as-Markov-chain analyses and in MCMC convergence proofs.

Information-theoretic arguments. Data processing inequalities, Fano's inequality, mutual information bounds. Used in lower bounds on learning and in information-bottleneck analyses.

Random matrix theory. Concentration of eigenvalues, Marchenko–Pastur and Tracy–Widom laws, spectral universality. Used in analyses of high-dimensional features, kernel methods, and neural network initialisation.

Mean-field and propagation of chaos. Replacing many-particle interactions with their averages, with rigorous control of the approximation error. Used in mean-field neural network analyses.

PDE and SDE methods. Treating training as evolution of a measure governed by partial or stochastic differential equations. Used in mean-field analyses, in diffusion models, in continuous-time SGD analyses.

Tensor decomposition arguments. Method of moments, identifiability via tensor decompositions. Used in latent variable models and in some neural network identifiability results.

Game-theoretic equilibrium arguments. Minimax theorems, no-regret learning, fictitious play. Used in GANs, in robust optimisation, in multi-agent settings.

Probabilistic method. Existence proved by computing that a random construction satisfies the requirement with positive probability. Used in expressivity lower bounds and in some constructive results.

The analytical move at this layer is to locate the moment of truth — the single step where the substance of the result is established rather than the algebra worked out — and to identify the machinery on which it rests. A paper is technically novel to the extent that it either introduces machinery beyond the standard set for its sub-literature or applies the standard set to a regime where it was not previously known to work. Most papers in mature sub-literatures fall in the second category; recognising this is part of the analyst's job, not a criticism in itself.

Layer 5: The Empirical–Theoretical Bridge

The fifth layer assesses the relationship between the formal claims and the empirical regime in which the paper's contribution is deployed. The primitives at this layer come in two kinds: the recognised types of gap between theory and practice, and the recognised moves for bridging or assessing the gap.

Types of gap

Idealisation gap. The theorem holds in an infinite limit — width, data, time, precision — while empirical work is finite.

Architectural gap. The theorem is for a simplified model (linear, shallow, single layer, specific activation) while empirical work uses a richer one.

Algorithmic gap. The theorem is for gradient flow, full-batch gradient descent, or exact Bayesian inference; empirical work uses SGD with finite step sizes, mini-batches, or variational approximation.

Distributional gap. The theorem assumes i.i.d., bounded, or specifically-distributed data; empirical work has distribution shift, heavy tails, or unspecified structure.

Measurement gap. The theorem speaks to one quantity (e.g. empirical risk) while empirical work measures another (e.g. zero-shot benchmark performance, a downstream metric).

Tightness gap. The theorem gives a loose bound while empirical performance is substantially better.

Scope gap. The theorem is about a specific case (single dataset, single architecture, single regime) while the paper's claim is general.

Bridge moves

Non-vacuity verification. Computing the bound at realistic scale and checking that its non-trivial regime overlaps with the empirical regime.

Idealisation acknowledgment. Explicitly stating which idealisation is in force and what it excludes.

Quantitative prediction matching. Testing whether the theory's predicted rates or scaling are observed.

Numerical instantiation of bounds. Actually computing constants and rates on realistic problems rather than reporting only asymptotics.

Assumption checking. Empirically verifying that the theorem's conditions hold in the regime of interest.

The analytical move at this layer is to identify the type of gap (or types — many papers have several) and to assess which bridge moves the paper makes. Some sub-literatures have established norms — non-vacuity in generalisation bounds since Dziugaite and Roy (2017), idealisation acknowledgment in NTK work since the feature-learning critiques of Chizat, Oyallon and Bach (2019) and Yang and Hu (2021) — and the paper's bridge-handling can be evaluated against these norms. A distinctive feature of contemporary AI is that empirical results regularly outrun theoretical predictions while theoretical results regularly describe regimes that do not correspond to deployed systems; the analyst should resist the temptation to treat theory and experiment as mutually supportive when they may be operating on disjoint objects.

Layer 6: Methodological Audit

The sixth layer checks for the recurring methodological failure modes of AI mathematics. The primitives at this layer are the recognised failure modes themselves.

Vacuous bounds. A generalisation bound that exceeds unity, or a sample complexity bound whose constants render the result useless at any realistic scale. A bound is informative only if its non-trivial regime overlaps with the empirical regime.

Hidden assumptions. An assumption introduced casually in a lemma, or labelled a "regularity condition," that turns out to be doing all the work in the proof. The audit move is to reverse the dependency graph and ask which assumption, if removed, would break the result.

Scope creep. The gap between what the theorem proves and what the introduction, abstract, and conclusion claim. The audit move is to record, separately, what is proven, what is claimed in the abstract, and what is suggested in the conclusion. The differences between these registers are the scope-creep budget.

Notational sleight-of-hand. Mathematical notation deployed without ensuring that the notated quantities are well-defined under the paper's assumptions. The information-bottleneck literature's $I(X; T)$ for deterministic networks is the canonical example.

Selective formalisation. Formalising the easy parts and leaving the conceptually difficult parts informal. The audit move is to ask whether the formal apparatus addresses the genuinely hard moves of the argument or only the routine ones.

Reference inflation. Citing a deep theoretical result as if it were instantiated in the present work when it is in fact only invoked. The audit move is to check whether cited theorems are actually used to derive a consequence in the paper.

Cherry-picked baselines. Empirical comparisons that select advantageous baselines, datasets, or hyperparameters. Not strictly a mathematical failure mode but often co-occurring with mathematical ones.

Definitional drift. A concept defined one way in the formal setting and used another way in the discursive setting. Common with terms such as "convergence", "generalisation", "robustness", "alignment".

Confused causality. Inferring a causal relation from a correlational result, or from a result whose formal statement does not license the causal reading.

Goalpost movement. When initial experimental results do not support the theoretical claim, the claim is restated to fit the results rather than the experiment redesigned to test the original claim.

Overstated novelty. Presenting as new what is a routine extension or rediscovery, often by citing only the most recent prior work rather than the original source.

Metric gaming. Optimising a measurable proxy in a way that disconnects it from the underlying quantity of interest, with the formal claim made about the proxy and the substantive claim made about the underlying quantity.

The analytical move at this layer is to run each failure mode as a check against the paper. A paper may pass all earlier layers and still fail at this one; conversely, a paper may exhibit one or two failure modes in minor ways and still make a real contribution. The audit is calibration, not condemnation.

A Compact Reading Protocol

In practical use the framework reduces to a sequence of analytical moves applied in order.

First, identify the mathematical object (Layer 1 primitives). Second, locate the theoretical frame (Layer 2 primitives) and note which frames are absent. Third, classify each claim by type, strength, conditions, and scope (Layer 3 primitives, along four axes). Fourth, identify the moment of truth in each proof and the machinery on which it rests (Layer 4 primitives). Fifth, identify the type of empirical–theoretical gap and assess what bridge moves the paper makes (Layer 5 primitives). Sixth, run the methodological audit against each recognised failure mode (Layer 6 primitives).

A complete pass produces a structured record: the object, the frame, the taxonomy of claims, the proof architecture, the bridge assessment, and the audit. This record can be retained as a research note and aggregated across papers to identify patterns in a research programme, in a research group, or in the literature as a whole

The framework is deliberately frame-agnostic and deliberately critical. It does not commit to a single theoretical tradition but asks where any given paper sits within the recognised primitives at each layer. Used carefully, it allows a reader to extract from any AI paper a precise account of what is being studied, how it is being framed, what is being claimed, how those claims are established, what relationship they bear to practice, and where the methodological vulnerabilities lie.

We now explore each of these six layers in detail

Exploring layers/primitives in detail

Layer 1 – Mathematical Objects

What mathematical object is actually being studied?

Every AI paper is ultimately about **something mathematical**. Although authors may describe models, algorithms, agents, or systems, these are often informal descriptions. The first task of the mathematical analyst is to identify the precise mathematical object that lies underneath.

This object might be a function, a probability distribution, an optimisation trajectory, a geometric space, an information-theoretic quantity, a game-theoretic equilibrium, or another well-defined mathematical entity. Identifying the object establishes exactly what the paper is analysing and provides the foundation for everything that follows.

The key insight is that **the same AI system can be viewed through different mathematical objects**. For example, a neural network may be regarded as a function, a trajectory through parameter space during training, a probability distribution over parameters, or a collection of computational circuits. Each viewpoint highlights different properties and supports different kinds of mathematical analysis.

Before asking whether a theorem is correct, or whether an algorithm performs well, the analyst should first ask:

"What is the mathematical object that this paper is really studying?"

Correctly identifying that object is the first step toward understanding the mathematics of the paper.

1. Empirical Risk Minimiser (ERM)

What is it?

Imagine a student taking an exam. The "risk" is simply how many questions they get wrong. During training, an AI model also tries to reduce its mistakes. The empirical risk minimiser is the model that makes the fewest mistakes on the training data.

Mathematically,

Risk = Average Loss over the training examples.

The algorithm searches through a huge collection of possible models (the hypothesis class) and selects the one with the smallest empirical risk.

Why is it important?

Nearly all supervised machine learning begins here. Linear regression, logistic regression, neural networks and decision trees all minimise some form of empirical risk.

Typical research question

Does the model that performs well on the training data also perform well on unseen data?

This naturally leads into statistical learning theory.

2. The Trained Network as a Function

After training, we usually stop thinking about millions of parameters and instead think of the network as one giant mathematical function.

Instead of Weights → Output we simply write $f(x)$ where the function maps inputs to predictions.

Researchers then ask questions like

- Is the function smooth?
- Is it stable?
- How sensitive is it to small perturbations?
- How complex is it?

This viewpoint leads naturally to generalisation theory and robustness.

3. Parameter Trajectory During Training

Training is not just about the final model. The entire learning process is a journey through an enormous parameter space. Imagine hiking through mountains trying to reach the lowest valley.

Every gradient descent step changes the model slightly.

The sequence

$\theta_0 \to \theta_1 \to \theta_2 \to \dots$

is called the parameter trajectory.

Researchers ask

- Why does SGD avoid bad minima?
- Why do networks suddenly improve?

- Why do different random initialisations end in similar places?

This connects AI with dynamical systems.

4. Distribution over Parameters

Instead of following one model, we can imagine an entire population of possible models.

Rather than saying "The parameters equal θ" we instead say "The parameters follow a probability distribution." Now learning becomes the evolution of an entire distribution rather than one point.

This viewpoint appears in

- Bayesian neural networks
- Mean-field theory
- Particle methods

It becomes especially important for understanding extremely large neural networks.

5. Network-Induced Kernels

One of the biggest surprises in modern AI is that some neural networks behave almost exactly like kernel machines. Instead of analysing neurons directly, we analyse the kernel generated by the network. This allows researchers to use decades of beautiful mathematics from kernel methods. The famous example is the Neural Tangent Kernel (NTK).

Research questions include

- Why do very wide networks behave almost linearly?
- Why can kernels explain early training?

6. Information-Theoretic Representations

Here the network is viewed as an information processing system.

Each hidden layer stores information about the input.

Researchers ask

- How much information survives?
- What information is discarded?
- Which information is useful?

Information theory provides tools like

- Entropy
- Mutual Information
- Channel Capacity

The Information Bottleneck hypothesis argues that good representations retain only information relevant for prediction.

7. Loss Landscape

Instead of studying the network itself, we study the shape of the optimisation problem. Imagine flying over mountains. High mountains represent large loss. Valleys represent good solutions.

Researchers investigate

- Sharp minima
- Flat minima
- Saddle points
- Connected valleys

This explains why optimisation often succeeds despite enormous complexity.

8. Data Distribution

Rather than analysing the model, some papers analyse the data itself. Where do the observations come from? Different generative models assume different mathematical structures.

Examples include

- Gaussian distributions
- Diffusion processes
- Autoregressive models
- Normalising flows

Understanding the data distribution is central to generative AI.

9. The Optimiser

Sometimes the optimiser itself becomes the mathematical object. Rather than asking "What did SGD produce?" we ask "Why does SGD behave the way it does?"

Researchers study

- SGD noise
- Momentum

- Adam
- Learning rate schedules
- Implicit regularisation

The optimiser becomes a stochastic dynamical system.

10. Mechanistic Computational Objects

This is one of the newest research areas.

Instead of treating the neural network as a black box, researchers identify internal computational structures such as

- Attention heads
- Features
- Circuits
- Induction heads
- Residual stream directions

The goal is to understand **how** a model actually reasons internally.

Mechanistic interpretability attempts to reverse-engineer the algorithms learned by neural networks.

11. Scaling Laws

Instead of analysing one model, researchers analyse entire families of models. As model size, compute and data increase, many quantities follow remarkably simple power laws.

Typical questions include

- How much performance improves if parameters double?
- How much data is needed?
- When does additional compute stop helping?

Scaling laws have become fundamental for modern foundation models.

12. Game-Theoretic Equilibria

Some AI systems involve multiple intelligent agents.

Examples include

- GANs
- Multi-agent reinforcement learning
- Auctions

- Self-play

Researchers study equilibrium points where no participant has an incentive to change strategy.

Game theory provides concepts such as

- Nash equilibrium
- Minimax optimisation
- Regret minimisation

13. Bayesian Posteriors

Instead of producing a single answer, Bayesian AI maintains uncertainty.

Learning updates beliefs using Bayes' theorem.

Prior → Data → Posterior

This framework naturally answers

- How confident is the model?
- How uncertain are predictions?
- How should evidence change beliefs?

Layer 2 – Theoretical Frame

A useful way to think about Layer 2 is that each theoretical frame asks a fundamentally different question about the same AI system:

Statistical Learning - Will it generalise?
Optimisation - Can we train it?
Information Theory - What information is represented?
Approximation Theory - Can it represent the target function?
Dynamical Systems - How does learning evolve over time?
Geometry & Topology - What is the shape of the representation space?
Probability - How should we model uncertainty?
Computational Complexity - Can it be computed efficiently?
Statistical Physics - How does collective behaviour emerge?
Bayesian Inference - How should beliefs change with evidence?
Causal Inference - What causes what?

Game Theory - What happens when multiple intelligent agents interact?

Layer two addresses: "What mathematical question is this paper trying to answer?"

Once you recognise the theoretical frame, you immediately understand what kinds of claims the paper is equipped to make—and just as importantly, what kinds of claims it is not equipped to make.

1. Statistical Learning Theory

What is it?

Statistical learning theory studies why models that perform well on training data also perform well on unseen data. It is concerned with **generalisation**.

Rather than asking "Can we fit the data?", it asks "Can we trust the model on new data?"

Its central concern is the relationship between:

- Training error
- Test error
- Model complexity
- Amount of data

Why is it important?

A model that memorises the training data is of little practical use. Statistical learning theory provides mathematical tools for understanding when learning is genuine rather than memorisation.

This field introduced ideas such as:

- PAC Learning
- VC Dimension
- Rademacher Complexity
- Algorithmic Stability
- PAC-Bayes

Typical research questions

- Why does deep learning generalise?
- How much data is enough?
- Which models are likely to overfit?
- How does model complexity affect prediction accuracy?

2. Optimisation Theory

What is it?

Learning is fundamentally an optimisation problem.

Training a neural network means searching through millions—or billions—of possible parameter values to find one that minimises the loss function.

Optimisation theory studies how this search works.

Why is it important?

Without optimisation, modern AI would not exist.

Researchers study questions such as:

- Will gradient descent converge?
- How quickly?
- Does it reach a good solution?
- Can it escape saddle points?
- Does Adam behave differently from SGD?

Typical research questions

- Why does stochastic gradient descent work so well?
- Which optimiser converges fastest?
- Why are some optimisation problems easier than others?

3. Information Theory

What is it?

Information theory studies how information is represented, transmitted and compressed. Originally developed for communication systems, it has become one of the most influential mathematical languages in AI.

Its fundamental quantities include:

- Entropy
- Mutual Information
- KL Divergence
- Channel Capacity

Why is it important?

Many AI problems involve extracting useful information while discarding irrelevant information.

Information theory provides precise ways of measuring this.

Examples include:

- Representation learning
- Information Bottleneck
- Variational Autoencoders
- Language models

Typical research questions

- How much information is stored inside hidden layers?
- Which features matter?
- What information is lost during learning?

4. Approximation Theory

What is it?

Approximation theory studies how well one mathematical function can approximate another. A neural network is ultimately a function approximator.

The key question is: *"Can this architecture represent the function we want to learn?"*

Why is it important?

Before worrying about optimisation or generalisation, we first ask whether the model is expressive enough.

Approximation theory explains ideas such as:

- Universal Approximation Theorem
- Depth versus width
- Expressive power
- Approximation rates

Typical research questions

- Can transformers approximate arbitrary functions?
- Is depth more important than width?
- How many neurons are needed?

5. Dynamical Systems

What is it?

Dynamical systems theory studies how systems evolve over time. Instead of viewing training as solving an optimisation problem, we view it as a continuously evolving process. The trajectory itself becomes the object of study.

Why is it important?

Many surprising behaviours emerge only when we study learning as a dynamic process.

Researchers investigate:

- Stability
- Oscillations
- Attractors
- Phase transitions
- Edge of stability

Typical research questions

- Why do neural networks suddenly improve?
- Why does SGD oscillate?
- What determines convergence?

6. Geometry and Topology

What is it?

Geometry studies shape. Topology studies the deeper structure of spaces that remains unchanged under continuous deformation.

In AI, high-dimensional data often has rich geometric structure.

Why is it important?

Modern AI is fundamentally geometric.

Examples include:

- Word embeddings
- Latent spaces
- Manifold learning
- Representation spaces
- Persistent homology

Researchers ask how data is organised geometrically.

Typical research questions

- Why do embeddings cluster?
- What is the geometry of latent space?
- Does curvature affect learning?

7. Probability Theory

What is it?

Probability theory studies randomness and uncertainty.

Almost every AI model assumes that data is generated randomly from some unknown probability distribution.

Why is it important?

Without probability there is no modern statistics or machine learning.

Probability underpins:

- Sampling
- Random variables
- Bayesian methods
- Concentration inequalities
- Stochastic optimisation

Typical research questions

- How likely is this prediction?
- How uncertain is the model?
- How does randomness affect training?

8. Computational Complexity

What is it?

Computational complexity studies the resources required to solve problems. A mathematically elegant algorithm may still be useless if it takes centuries to run. Complexity asks whether a computation is practical.

Why is it important?

Large AI models consume enormous computational resources.

Researchers analyse:

- Time complexity
- Memory complexity
- Communication cost
- Query complexity

Typical research questions

- Can this algorithm scale?
- What is the computational bottleneck?
- Is there a faster algorithm?

9. Statistical Physics

What is it?

Statistical physics explains how simple microscopic interactions give rise to complex collective behaviour. Originally developed for atoms, it now helps explain large neural networks.

Why is it important?

Very large AI systems often display surprising emergent behaviour.

Statistical physics provides mathematical tools for analysing:

- Phase transitions
- Free energy
- Spin glasses
- Emergence

Many ideas about deep learning landscapes originated here.

Typical research questions

- Why do large models suddenly acquire new capabilities?
- Why are loss landscapes surprisingly smooth?
- How does collective behaviour emerge?

10. Bayesian Inference

What is it?

Bayesian inference views learning as updating beliefs in light of evidence.

Rather than finding one "best" model, it maintains a probability distribution over many possible models. Learning is simply repeated application of Bayes' theorem.

Why is it important?

Bayesian methods naturally represent uncertainty.

This is especially valuable in:

- Scientific AI
- Medical diagnosis
- Robotics
- Decision making

Typical research questions

- How should evidence change our beliefs?
- How uncertain is the prediction?
- Which hypothesis is most plausible?

11. Causal Inference

What is it?

Most machine learning discovers correlations. Causal inference attempts to discover cause-and-effect relationships. It asks not merely: *"What happens together?"* but *"What would happen if we intervened?"*

Why is it important?

Many important decisions require causal reasoning rather than prediction.

Examples include:

- Healthcare
- Public policy
- Economics
- Scientific discovery

Key ideas include:

- Structural Causal Models
- Directed Acyclic Graphs (DAGs)
- Do-calculus
- Counterfactual reasoning

Typical research questions

- Will this treatment improve patient outcomes?
- What caused the failure?
- What would happen if we changed this variable?

12. Game Theory

What is it?

Game theory studies situations where multiple intelligent decision-makers interact.

Each participant adapts to the others. Learning therefore becomes a strategic process rather than an isolated optimisation problem.

Why is it important?

Many modern AI systems involve interacting agents.

Examples include:

- GANs
- Multi-agent reinforcement learning
- Auctions
- Autonomous vehicles
- Markets

Researchers analyse concepts such as:

- Nash equilibrium
- Regret minimisation
- Mechanism design
- Stackelberg games

Typical research questions

- Will the agents reach equilibrium?
- Can cooperation emerge?
- What strategy is optimal when others are learning too?

Layer 3 – Claim Structure

Layer 3 is where the framework becomes genuinely analytical rather than descriptive. Layers 1 and 2 identify **what** the paper studies and **which mathematical language** it uses. Layer 3 asks:

> **"Exactly what is the paper claiming?"**

Many AI papers appear impressive because readers never separate different kinds of claims. A convergence theorem is not a generalisation theorem. An approximation theorem is not an optimisation theorem. A lower bound is not a practical performance guarantee. Layer 3 teaches the reader to classify claims precisely.

Every mathematical claim can be analysed along **four dimensions**:

1. **Type of claim**
2. **Strength of claim**
3. **Conditions required**
4. **Scope of the claim**

Thinking this way prevents us from confusing what the theorem actually proves with what the authors suggest it proves.

Layer 3 teaches us to read every theorem using four simple questions:

- What kind of mathematical claim is being made?
- Strength: How strong is the guarantee?
- Conditions What assumptions are required?
- Scope How far can the conclusion legitimately be extended?

One of the biggest mistakes when reading AI papers is to focus only on the theorem itself. An experienced mathematical reader pays equal attention to what is claimed, how strong the claim is, what assumptions make it true, and where its validity ends. Those four questions form the heart of Layer 3 and are what distinguish a careful mathematical analysis from a superficial reading.

Part 1 – Types of Mathematical Claims

1. Existence

What is it?

An existence theorem proves that something exists without necessarily telling us how to find it.

For example,

"There exists a neural network capable of approximating every continuous function."

The theorem guarantees the existence of such a network.

It does **not** tell us

- how to build it,
- how large it is,
- how long training takes,
- or whether SGD will ever find it.

Why is it important?

Many foundational AI results begin with existence.

Examples include

- Universal Approximation Theorem
- Nash Equilibrium
- Fixed Point Theorems

Typical research question Does such a model exist?

2. Convergence

What is it?

Convergence concerns what happens as an algorithm continues running.

Does it eventually reach the desired solution?

For example, Gradient Descent may converge toward a local minimum; A reinforcement learning algorithm may converge toward an optimal policy.

Why is it important?

Training algorithms are iterative. Without convergence, training could oscillate forever or diverge completely.

Typical research questions

- Will SGD converge?

- How quickly?
- Under what conditions?

3. Generalisation

What is it?

Generalisation asks whether performance on unseen data will resemble performance on the training data.

This is arguably the central question of machine learning. A model may achieve 99% training accuracy but only 55% test accuracy. Generalisation theory attempts to explain why.

Why is it important?

Ultimately AI systems are deployed on new data, not the training data. Good generalisation distinguishes learning from memorisation.

Typical research questions

- Why do large neural networks generalise?
- How can overfitting be prevented?

4. Sample Complexity

What is it?

Sample complexity measures how much data is required to learn successfully.

Instead of asking *"Can we learn?"* it asks *"How many examples are needed?"*

Why is it important?

Data collection is often expensive. Knowing that only a few hundred examples are needed instead of millions has enormous practical value.

Typical research questions

- How many labelled images are required?
- How much data is sufficient?

5. Computational Complexity

What is it?

A problem may be mathematically solvable but computationally impossible.Computational complexity studies the resources required to solve problems.

Researchers analyse

- running time,
- memory,
- communication,
- query complexity.

Why is it important?

Modern AI operates at enormous scale.

An algorithm that is theoretically correct but computationally infeasible is often of little practical use.

Typical research questions

- Can this algorithm scale?
- Is there a faster algorithm?

6. Expressivity

What is it?

Expressivity measures what kinds of functions a model can represent. Before training even begins we ask *"Is this architecture capable of representing the desired function?"*

Why is it important?

A model cannot learn something it cannot represent.

Expressivity underlies

- Universal Approximation
- Transformer theory
- Graph Neural Networks
- Attention mechanisms

Typical research questions

- Are transformers universal approximators?
- Does depth increase expressive power?

7. Identifiability

What is it?

Sometimes different parameter values produce exactly the same observable behaviour. Identifiability asks whether the true underlying parameters can be uniquely recovered from data.

Why is it important?

Many scientific applications require discovering hidden causes rather than simply making predictions.

Examples include

- latent variable models,
- causal discovery,
- topic models,
- tensor decomposition.

Typical research questions

- Can the true parameters be recovered?
- Are multiple explanations equally valid?

8. Stability and Robustness

What is it?

A robust model changes only slightly when its inputs change slightly. Stability studies how sensitive the model is to perturbations.

Why is it important?

Real-world data is noisy. AI systems must tolerate

- measurement errors,
- adversarial attacks,
- distribution shifts.

Typical research questions

- Is the classifier robust?
- How sensitive are predictions?

9. Approximation Rate

What is it?

Approximation theory often asks not merely *"Can the model approximate?"* but *"How quickly does the approximation improve?"* As more neurons, layers or samples are added, approximation error usually decreases. Approximation rate measures how fast.

Why is it important?

Different architectures improve at different speeds.

Typical research questions

- Does error decrease exponentially?
- Does adding layers help more than adding neurons?

10. Lower Bounds and Impossibility

What is it?

Some theorems prove that something cannot be done. These are often more powerful than positive results. For example "No algorithm can achieve better than..." Such results define the fundamental limits of learning.

Why is it important?

Lower bounds tell researchers when to stop searching for impossible algorithms.

Typical research questions

- What is the minimum number of samples?
- Is perfect learning impossible?

11. Equivalence and Reduction

What is it?

Sometimes two seemingly different problems are mathematically identical. Alternatively one problem may reduce to another already understood problem.

Why is it important?

Reduction allows decades of previous mathematics to become immediately useful.

Typical research questions

- Is this optimisation problem equivalent to another?
- Can transformer attention be interpreted as kernel regression?

12. Calibration

What is it?

Calibration measures whether predicted probabilities match reality. Suppose a weather model predicts 80% chance of rain 100 times. If it is well calibrated, it should rain approximately 80 times.

Why is it important?

Modern AI often makes probabilistic predictions.Decision-making depends not only on accuracy but also on confidence.

Typical research questions

- Can we trust model confidence?
- Are prediction probabilities meaningful?

Part 2 – Strength of the Claim

Not all theorems are equally strong.

Every theorem should be judged along several dimensions.

Worst-case vs Average-case

Worst-case guarantees work under every possible situation. Average-case guarantees work only on typical situations. Worst-case results are stronger but often more conservative.

Finite-sample vs Asymptotic

Finite-sample theorems describe today's datasets. Asymptotic theorems describe behaviour as the amount of data approaches infinity. Finite-sample results are generally more useful in practice.

Distribution-free vs Distribution-dependent

Distribution-free results make almost no assumptions about the data. Distribution-dependent results assume specific statistical properties. Distribution-free theorems are more general. Distribution-dependent theorems are often much tighter.

High Probability vs Expectation

Some theorems guarantee "This succeeds with probability at least 99%." Others guarantee only "The average performance is good." High-probability results provide stronger practical guarantees.

Tight vs Loose Bounds

A bound may be mathematically correct but extremely conservative. For example, Actual error = 2% Theoretical upper bound = 97% The theorem is true but not very informative. Researchers increasingly ask whether bounds are non-vacuous.

Part 3 - Categories of Conditions

Every theorem rests on assumptions. These assumptions deserve as much attention as the conclusion.

Regularity Conditions

These ensure the mathematics behaves nicely.

Examples include

- continuity,
- differentiability,
- smoothness,
- Lipschitz continuity,
- boundedness.

Statistical Conditions

These concern the data.

Examples include

- IID sampling,
- bounded variance,
- sub-Gaussian noise,
- independence,
- mixing assumptions.

Structural Conditions

These describe the underlying problem.

Examples include

- sparsity,
- low rank,
- manifold hypothesis,
- separability,
- low intrinsic dimension.

Architectural Conditions

These describe the model.

Examples include

- transformer,
- CNN,
- network width,
- activation function,
- initialisation.

Algorithmic Conditions

These describe the training procedure.

Examples include

- SGD,
- Adam,
- learning rate,
- batch size,
- regularisation,
- number of iterations.

Part 4 - Scope Gaps

Perhaps the most valuable part of Layer 3 is recognising when authors quietly move from a narrow theorem to a broad claim.

Algorithmic Idealisation

The theorem assumes perfect gradient flow.

The experiments use noisy SGD.

The theorem therefore does not exactly describe the experiment.

Architectural Idealisation

The proof analyses a simplified network.

The experiments use GPT-scale transformers.

The mathematics applies only indirectly.

Limit-Taking

Many AI proofs assume

- infinite width,
- infinite data,
- infinite training time.

Real systems are finite.

The analyst should always note this gap.

Distributional Simplification

The theorem assumes

Gaussian IID data.

The experiments use internet-scale data.

Again, there is a gap between mathematics and reality.

Layer 4 – Proof Architecture

The first three layers ask:

- What is the mathematical object?
- Which theoretical frame is being used?
- What is being claimed?

Layer 4 asks a different question:

"How is the claim actually proved?"

Rather than checking every line of mathematics, we identify the proof's architecture.

Just as engineers recognise arches, trusses and suspension bridges, mathematicians recognise recurring proof techniques.

Once you recognise the architecture, many papers become much easier to read.

Layer 4 teaches you to recognise the standard toolbox of mathematical AI. Each proof technique is like a specialised engineering tool, suited to a particular class of problems

- Concentration Inequalities: Random quantities stay close to their averages.
- Covering & Chaining: Approximate infinitely many functions with a manageable finite set.
- Symmetrisation: Replace a hard probabilistic problem with an equivalent, easier one.
- Coupling: Analyse two random processes together.
- Martingales: Exploit the structure of fair stochastic processes.
- Fixed-Point Theorems: Prove stable solutions or equilibria exist.
- Variational Principles: Recast problems as optimisation.
- Lyapunov Functions: Show stability through a quantity that always decreases.
- Markov Chains: Study memoryless stochastic evolution.
- Information-Theoretic Arguments: Use limits imposed by information itself.
- Random Matrix Theory: Understand the collective behaviour of large matrices.
- Mean-Field Theory: Replace many interacting components with their average behaviour.
- PDE/SDE Methods: Model learning as continuous evolution.
- Tensor Decomposition: Recover hidden structure from high-dimensional data.
- Game-Theoretic Arguments: Analyse strategic interaction among agents.
- Probabilistic Method: Prove existence by showing random constructions succeed.

The key insight of Layer 4 is that experienced researchers rarely memorise every proof. Instead, they learn to recognise these recurring proof architectures. Once you can identify the underlying technique, you can often understand the essence of a paper long before you've worked through every line of mathematics.

1. Concentration Inequalities

What is it?

One of the biggest surprises in probability is that averages are often remarkably stable.

Although individual observations may vary wildly, their average usually stays close to its expected value. Concentration inequalities make this intuition precise. They provide mathematical guarantees that random quantities remain close to their expected values.

Famous examples include

- Hoeffding's Inequality
- Bernstein's Inequality
- McDiarmid's Inequality
- Bennett's Inequality
- Talagrand's Inequality

Why is it important?

Machine learning relies heavily on finite samples. We never observe the entire population. Concentration inequalities justify using a sample to estimate the behaviour of the whole distribution. Without them, statistical learning theory would largely collapse.

Typical research questions

- Why does training error estimate test error?
- How reliable is empirical risk?
- How quickly do averages converge?

Intuition

Imagine flipping a fair coin. After ten flips, you might obtain 7 Heads 3 Tails.

After one million flips, the proportion of heads is extremely close to 50%. Concentration inequalities quantify exactly how unlikely large deviations become.

2. Covering and Chaining

What is it?

Many proofs must analyse an enormous—sometimes infinite—collection of functions. That seems impossible. Covering arguments replace this infinite collection with a carefully chosen finite approximation. Instead of studying every point on a sphere, we study enough representative points to approximate the whole sphere. Chaining improves this approximation by building progressively finer coverings.

Why is it important?

Many generalisation bounds depend on controlling extremely complicated hypothesis classes. Covering numbers provide a measure of model complexity.

Typical research questions

- How large is the hypothesis class?
- How complex is this family of neural networks?
- Can we bound the worst-case prediction error?

Intuition

Imagine estimating the shape of the Earth. Instead of measuring every square metre, you measure carefully selected locations. If they are sufficiently dense, you can reconstruct the entire surface.

3. Symmetrisation

What is it?

Symmetrisation is a clever probabilistic trick. Instead of analysing complicated dependence on unknown data distributions, we compare the observed data with an independent copy of itself. This produces a much simpler mathematical problem.

Why is it important?

Many generalisation proofs begin with a difficult expectation. Symmetrisation converts it into something that concentration inequalities can handle. It is one of the most common techniques in statistical learning theory.

Typical research questions

- How large is the generalisation gap?
- Can empirical performance estimate population performance?

Intuition

Suppose two students independently solve the same exam. Comparing their answers often reveals the underlying variability more clearly than studying one student's paper alone.

4. Coupling Arguments

What is it?

A coupling constructs two random processes on the same probability space. Instead of studying each process independently, we let them evolve together. Their distance becomes much easier to analyse.

Why is it important?

Coupling provides elegant proofs of convergence.

It appears in

- Markov chains
- Stochastic processes
- Sampling algorithms
- Stability analyses

Typical research questions

- How quickly do two learning processes become similar?
- When does a stochastic algorithm stabilise?

Intuition

Imagine two hikers starting from different locations. Instead of following separate maps, we tie them together with a rope. Studying the rope's length tells us how their paths converge.

5. Martingale Arguments

What is it?

A martingale is a mathematical model of a fair game. Given everything known so far, the expected future value equals the current value. Martingales have extraordinarily powerful convergence theorems.

Why is it important?

Many online learning algorithms naturally form martingales.They allow researchers to prove regret bounds and convergence guarantees.

Typical research questions

- Does online learning converge?
- How large can cumulative error become?

Intuition

A gambler playing a perfectly fair game cannot systematically expect to become richer or poorer. Martingale mathematics formalises this idea.

6. Fixed-Point Theorems

What is it?

A fixed point is a value that remains unchanged by a transformation. Mathematically, $f(x) = x$ Many existence proofs reduce to showing that a fixed point must exist. Important examples include

- Banach Fixed Point Theorem
- Brouwer Fixed Point Theorem
- Schauder Fixed Point Theorem
- Kakutani Fixed Point Theorem

Why is it important?

Fixed-point theorems prove the existence of

- equilibria,
- optimal policies,
- stable solutions.

Typical research questions

- Does equilibrium exist?
- Will iterative updates eventually stabilise?

Intuition

Imagine stirring coffee.At least one tiny particle eventually ends up exactly where it started. That remarkable fact follows from Brouwer's Fixed Point Theorem.

7. Variational Principles

What is it?

Many difficult problems become easier if rewritten as optimisation problems. Instead of solving equations directly,we search for the function that minimises (or maximises) some objective.

Why is it important?

Modern AI is built around optimisation.

Variational methods underpin

- Variational Autoencoders
- Bayesian inference
- Energy-based models

Typical research questions

- Which distribution best explains the data?
- Which function minimises prediction error?

Intuition

Nature often finds the path of least action.Variational methods exploit exactly this principle.

8. Lyapunov Function Arguments

What is it?

A Lyapunov function is a mathematical measure of progress. If the quantity continually decreases,the system is moving toward stability.

Why is it important?

Rather than solving complicated differential equations, researchers simply show that one carefully chosen quantity always decreases.

Typical research questions

- Will learning stabilise?
- Is the equilibrium stable?

Intuition

Imagine rolling a marble into a bowl. Potential energy continually decreases until the marble comes to rest. The potential energy acts like a Lyapunov function.

9. Markov Chain Analysis

What is it?

A Markov chain is a process whose future depends only on its current state.

Researchers analyse

- stationary distributions,
- mixing times,
- convergence rates,
- spectral gaps.

Why is it important?

Many optimisation algorithms behave approximately like Markov chains.

So do

- MCMC algorithms,
- reinforcement learning,
- stochastic optimisation.

Typical research questions

- How long before randomness settles?
- Has the algorithm converged?

Intuition

Weather tomorrow depends far more on today's weather than last month's. That "memoryless" property defines a Markov process.

10. Information-Theoretic Arguments

What is it?

Instead of analysing algorithms directly, we analyse how much information is available. Information inequalities often establish surprisingly strong lower bounds.

Important tools include

- Data Processing Inequality
- Fano's Inequality
- Mutual Information Bounds

Why is it important?

Sometimes information itself limits what learning is possible. No algorithm can recover information that simply is not present.

Typical research questions

- How much information is needed?
- Why can't any algorithm solve this problem better?

11. Random Matrix Theory

What is it?

Modern neural networks contain enormous matrices. Random Matrix Theory studies their collective behaviour.Instead of individual eigenvalues, it investigates the statistical behaviour of entire spectra.

Why is it important?

Large neural networks often display remarkably predictable spectral behaviour. Random Matrix Theory explains phenomena such as

- neural network initialisation,
- feature learning,
- covariance estimation.

Typical research questions

- Why do deep networks initialise well?
- What does the Hessian spectrum reveal?

12. Mean-Field and Propagation of Chaos

What is it?

Instead of analysing billions of interacting neurons individually,we study their average behaviour. As the network grows, individual interactions become increasingly well approximated by averages.

Why is it important?

Mean-field theory explains why extremely large neural networks often become mathematically simpler rather than more complicated.

Typical research questions

- What happens as width approaches infinity?
- Can one neuron represent the behaviour of many?

13. PDE and SDE Methods

What is it?

Some AI systems evolve continuously over time. Rather than studying individual parameter updates, we model learning using

- Partial Differential Equations (PDEs)
- Stochastic Differential Equations (SDEs)

Why is it important?

Continuous mathematics often reveals elegant structures hidden inside discrete optimisation algorithms.

Typical research questions

- How does a probability distribution evolve?
- Can training dynamics be modelled continuously?

14. Tensor Decomposition Arguments

What is it?

Many high-dimensional datasets naturally form tensors rather than matrices.Tensor decomposition attempts to separate these tensors into simpler components.

Why is it important?

Tensor methods can recover hidden variables that ordinary matrix methods cannot.

They appear in

- latent variable models,
- topic modelling,
- representation learning.

Typical research questions

- Can hidden structure be recovered?
- Are latent components unique?

15. Game-Theoretic Equilibrium Arguments

What is it?

When several learning agents interact, proofs often revolve around equilibrium. Researchers show that repeated strategic interaction converges toward stable behaviour.

Why is it important?

GANs, reinforcement learning and multi-agent AI all depend upon strategic interaction.

Typical research questions

- Does equilibrium exist?
- Will learning stabilise?
- Is one player guaranteed to improve?

16. The Probabilistic Method

What is it?

One of the most elegant techniques in modern mathematics.Instead of constructing an object directly, we show that a randomly chosen object has a positive probability of possessing the required property. If such an object occurs with positive probability, then it must exist.

Why is it important?

Many powerful existence proofs become remarkably simple using randomness.

Typical research questions

- Does such a network exist?
- Can this structure be constructed?
- Is this performance theoretically achievable?

Layer 5 – The Empirical–Theoretical Bridge

Layer 5 asks whether **the mathematics and the experiments are talking about the same system**.

The first four layers ask

- What mathematical object is being studied?
- Which mathematical framework is used?
- What claims are being made?
- How are those claims proved?

Layer 5 asks a different question:

"Does the mathematics actually explain the empirical results?"

This is not always true.

Some of the most successful AI systems have theories that explain only small parts of their behaviour.

Likewise, some beautiful theorems describe situations that never occur in practice.

Learning to recognise this gap is one of the hallmarks of a mature AI researcher.

Many frameworks stop after discussing the mathematics. But AI is unusual because **the mathematics and the engineering often live in different worlds**. A theorem may be perfectly correct and yet have very little to say about the model that is actually deployed. Conversely, a model may work extraordinarily well while our theory explains only a simplified version of it.

Layer 5 asks perhaps the most important question in modern AI:

"How well does the mathematics actually describe the system being built?"

This layer is about building—or evaluating—the bridge between theory and practice.

Perhaps the most important lesson of Layer 5 is that **a beautiful theorem and a successful experiment do not automatically validate one another**. They may be describing different objects, operating under different assumptions, or measuring different quantities. The role of the mathematical analyst is to determine how strong—or how fragile—the bridge is between the formal theory and the empirical evidence. That is one of the defining challenges of modern artificial intelligence research.

A useful way to summarise it is:

- Idealisation Gap: Is the mathematical world too idealised?
- Architectural Gap: Does the analysed model match the deployed model?
- Algorithmic Gap: Does the proof study the algorithm actually used?
- Distributional Gap: Do the data assumptions match reality?
- Measurement Gap: Are the theorem and the experiments measuring the same thing?
- Tightness Gap: Is the mathematical bound informative in practice?
- Scope Gap: Are the paper's conclusions broader than the theorem supports?

Good researchers do not stop at identifying these gaps. They try to close them through possible Bridge moves ie how theory relates to practice.

Non-Vacuity Verification: Check whether theoretical bounds are meaningful at realistic scales.

- Idealisation Acknowledgment: Clearly state simplifying assumptions and their limitations.
- Quantitative Prediction Matching: Compare theoretical predictions with observed empirical behaviour.
- Numerical Instantiation of Bounds: Replace asymptotic notation with concrete numerical estimates.
- Assumption Checking: Verify experimentally that the theorem's assumptions are approximately satisfied.

Part 1 – Types of Gap

1. Idealisation Gap

What is it?

Mathematics often studies idealised worlds. Real engineering rarely operates under those ideal conditions. A theorem might assume

- infinite training time,
- infinitely wide networks,
- exact arithmetic,
- unlimited data.

Real AI systems have

- finite GPUs,
- finite datasets,
- numerical errors,
- limited budgets.

Why is it important?

Idealisations make mathematics tractable. But they also limit what the theorem tells us about deployed systems. The analyst should always ask

"How close is reality to the mathematical ideal?"

Typical example

Neural Tangent Kernel theory assumes infinite-width networks. GPT models are enormous,but still finite. The theory is therefore an approximation, not an exact description.

2. Architectural Gap

What is it?

The mathematics analyses one architecture. The experiments use another. For example, the theorem proves results for shallow linear networks, while experiments use deep transformers with attention, residual connections and layer normalisation.

Why is it important?

Mathematical simplification is often necessary. However, the further the deployed architecture moves from the analysed architecture, the weaker the explanatory connection becomes.

Typical example

A proof for a single-layer network is often used to motivate intuition about deep learning. That intuition may be useful, but it is not the theorem.

3. Algorithmic Gap

What is it?

The theorem studies one optimisation algorithm. The implementation uses another.

Examples include

Theory:

- Gradient Flow
- Full Batch Gradient Descent
- Exact Bayesian Inference

Practice:

- SGD
- Adam
- Mini-batches
- Approximate inference

Why is it important?

Small algorithmic differences can produce dramatically different behaviour.The analyst should identify exactly where theory and implementation diverge.

Typical example

Many convergence proofs analyse gradient flow because differential equations are easier than stochastic optimisation. Industrial AI almost never trains that way.

4. Distributional Gap

What is it?

Mathematics usually assumes well-behaved probability distributions. Reality rarely cooperates.

The theorem might assume

- IID sampling,
- Gaussian noise,
- bounded variance.

Real-world datasets contain

- distribution shift,
- bias,
- missing data,
- heavy-tailed noise,
- changing environments.

Why is it important?

Many theorems depend critically upon statistical assumptions. If those assumptions fail, the theoretical guarantees may disappear.

Typical example

Medical datasets collected in one country often behave differently when deployed elsewhere.The mathematics may still be correct,but its assumptions no longer hold.

5. Measurement Gap

What is it?

The theorem proves something about one quantity. The experiments measure something else. For example, Theory vs practice can be seen with Empirical Risk vs Benchmark Accuracy Or Cross-Entropy Loss vs Human Preference Scores

Why is it important?

Researchers sometimes assume these quantities are interchangeable. Often they are only loosely related but not interchangeable.

Typical example

Reducing training loss does not necessarily improve downstream reasoning ability.

6. Tightness Gap

What is it?

The theorem provides a mathematically correct bound. Unfortunately, the bound may be far too loose to explain reality. For example, Theory: Error $\leq$ 95% Observed error: 2%. Thus, the theorem is true,but it explains almost nothing about actual performance.

Why is it important?

Many classical generalisation bounds are mathematically elegant but practically vacuous. Modern AI increasingly seeks **non-vacuous** theoretical guarantees.

7. Scope Gap

What is it?

The theorem proves something very specific. The paper discusses it as though it applies everywhere. For example, The theorem proves convergence for one architecture, on one dataset, under one optimiser. The discussion concludes "Deep learning converges."

Why is it important?

Scope expansion is extremely common. The analyst separates what is proven, from what is suggested.

Part 2 – Bridge Moves

Good papers recognise these gaps. Excellent papers actively try to bridge them.

1. Non-Vacuity Verification

What is it?

Instead of merely proving a bound, the authors calculate it using realistic numbers.

If the theoretical bound predicts Error ≤ 0.25 and observed error is 0.21 the observation is genuinely informative. Why is it important? Mathematics should illuminate practice. A theorem whose numerical values are meaningless contributes much less.

Typical research question

Does the theoretical guarantee actually hold at GPT scale?

2. Idealisation Acknowledgment

What is it?

Good researchers openly state ex "Our analysis applies only to convex objectives."

Why is it important?

Being explicit about assumptions helps readers judge applicability. It also identifies opportunities for future work.

Good scientific practice Never hide the simplifying assumptions. Explain exactly what they exclude.

3. Quantitative Prediction Matching

What is it?

A strong theory predicts measurable behaviour. Researchers then test whether experiments follow those predictions.

For example, Theory predicts Loss decreases proportional to $n^{-1/2}$. Experiments measure whether loss actually follows that scaling law.

Why is it important?

This is how theories become scientifically useful. They make predictions that experiments can confirm—or refute.

Typical research questions

Do observed scaling laws match theoretical scaling laws?

Does convergence occur at the predicted rate?

4. Numerical Instantiation of Bounds

What is it?

Rather than presenting only asymptotic notation, authors substitute realistic values. Instead of writing $O(\sqrt{n})$ they calculate the expected error for 10 million samples.

Why is it important?

Concrete numbers reveal whether a theorem is practically meaningful.

Typical example

A complexity theorem predicts training time.The authors estimate runtime for today's hardware.

5. Assumption Checking

What is it?

Every theorem has assumptions. Rather than assuming they hold, researchers verify them experimentally.

Examples include

- Is the data approximately IID?

- Is the network sufficiently wide?
- Are gradients actually smooth?
- Are Lipschitz assumptions approximately satisfied?

Why is it important?

A theorem is only as useful as its assumptions are realistic. Testing assumptions strengthens confidence that the mathematics applies to the empirical setting.

Layer 6 – Methodological Audit

Layer 6 is the culmination of the entire framework.

The first five layers teach you to understand a paper. Layer 6 teaches you to **critically evaluate** it. This layer is about developing good scientific judgement. Every mathematical paper has limitations, assumptions and trade-offs. The best researchers are those who can recognise both the strengths and the weaknesses of a piece of work.

Layer 6 therefore asks one final question: **"Can we trust the conclusions the paper draws?"**

A methodological audit is similar to the final inspection of a bridge before it opens to the public. The engineers have designed it. The mathematics appears correct. The calculations have been checked.

Now we ask: **"Where might this fail?"**

That is the purpose of Layer 6.

The framework identifies recurring failure modes that appear throughout AI research. Learning to recognise them is one of the defining skills of an experienced researcher.

1. Vacuous Bounds

What is it?

A theorem may provide a mathematically correct bound that is so loose it is practically useless. For example, suppose a paper proves Generalisation Error ≤ 1.8 But classification error can never exceed 1. The theorem is technically true but tells us nothing useful.

Why is it important?

Mathematics should provide insight, not merely correctness. A useful theorem narrows our uncertainty. A vacuous theorem leaves us almost exactly where we started.

Typical example

Many early deep learning generalisation bounds became enormous as network size increased. Although mathematically correct, they failed to explain why deep networks worked so well.

Audit question

Does the bound provide meaningful information at realistic scales?

2. Hidden Assumptions

What is it?

Sometimes the theorem appears powerful until one discovers a small assumption buried deep inside the proof. That assumption quietly carries almost all of the mathematical weight.

Why is it important?

The conclusion may sound general.

In reality, everything depends on one highly restrictive condition.

Typical example

A convergence theorem assumes

- convex loss,
- Lipschitz gradients,
- bounded variance,

yet the abstract simply states

"We prove convergence."

Audit question

Which assumption would cause the theorem to fail if removed?

That is usually the most important assumption.

3. Scope Creep

What is it?

The theorem proves one thing. The introduction claims something larger. The conclusion suggests something even larger. The scope gradually expands throughout the paper.

Why is it important?

Readers often remember the conclusion rather than the theorem. Scope creep creates an exaggerated impression of what has actually been established.

Typical example

The theorem proves - "A simplified transformer converges." The paper concludes "This explains transformer learning." Those are not the same claim.

Audit question

Compare three statements separately:

- What the theorem proves.
- What the abstract claims.
- What the conclusion implies.

The differences reveal the scope-creep budget.

4. Notational Sleight-of-Hand

What is it?

Mathematical notation can sometimes create an illusion of precision. A quantity is written using formal symbols, but under the stated assumptions it is not actually well defined. The notation looks rigorous even though the mathematics underneath is incomplete.

Why is it important?

Notation should clarify thinking. It should never replace thinking.

Typical example

The Information Bottleneck literature originally discussed mutual information in deterministic neural networks. Under deterministic mappings, this quantity is not always mathematically well defined. The notation appeared rigorous, but additional assumptions were required.

Audit question

Is every mathematical symbol actually well defined?

5. Selective Formalisation

What is it?

Researchers sometimes formalise the easy parts of an argument while leaving the genuinely difficult ideas informal. The mathematics is correct, but it addresses only the least controversial aspects.

Why is it important?

A long proof does not necessarily prove the central idea. Sometimes it proves only the convenient parts.

Typical example

A paper rigorously analyses optimisation, while treating representation learning entirely through intuition.

Audit question

Does mathematics address the paper's central scientific question?

6. Reference Inflation

What is it?

A paper cites powerful mathematical results to create an impression of theoretical depth, even though those results are never actually used.

Why is it important?

Citations should support the argument,not decorate it.

Typical example

A paper discusses PAC learning extensively, yet no PAC theorem appears in the actual proof.

Audit question

Which cited results genuinely contribute to the proof?

7. Cherry-Picked Baselines

What is it?

Empirical evaluation compares the proposed method against unusually weak competitors. The new method therefore appears stronger than it really is.

Why is it important?

Even excellent mathematics cannot compensate for unfair experiments. Scientific comparison requires strong and representative baselines.

Typical example

A new optimisation algorithm is compared only against SGD, while stronger modern optimisers are omitted.

Audit question

Have the strongest reasonable baselines been included?

8. Definitional Drift

What is it?

A word changes meaning during the paper. The mathematical definition differs from the informal discussion. Common examples include

- robustness,
- intelligence,
- alignment,
- convergence,
- generalisation.

Why is it important?

Readers assume terminology remains consistent. Changing definitions without notice creates confusion.

Typical example

Generalisation initially means low test error. Later it becomes strong reasoning ability. These are different concepts.

Audit question

Does every important term retain the same meaning throughout the paper?

9. Confused Causality

What is it?

A paper observes a correlation, then discusses it as though it demonstrates causation.

Why is it important?

Correlation alone rarely justifies causal conclusions. Additional assumptions or experiments are usually required.

Typical example

Models with larger context windows perform better. The paper concludes "Larger context windows cause better reasoning." Other explanations may exist.

Audit question

Does the evidence justify causal language?

10. Goalpost Movement

What is it?

The original hypothesis fails. Instead of redesigning the experiment,the hypothesis quietly changes.The revised hypothesis now matches the observed results.

Why is it important?

Scientific integrity requires testing hypotheses, not rewriting them afterwards.

Typical example

A paper initially claims improved accuracy.After seeing the results, the emphasis shifts to faster training instead.

Audit question

Did the evaluation remain faithful to the original research question?

11. Overstated Novelty

What is it?

A paper presents an incremental extension as though it were a major breakthrough. Often this happens because only recent literature is cited,while older foundational work is overlooked.

Why is it important?

Scientific progress builds upon previous discoveries. Accurate attribution strengthens rather than weakens new research.

Typical example

A "new" optimisation algorithm differs from an existing method only by a minor parameter choice.

Audit question

What is genuinely new, and what is an extension of existing work?

12. Metric Gaming

What is it?

Researchers optimise a measurable quantity that gradually becomes disconnected from the real objective. The benchmark improves, but the underlying capability does not.

Why is it important?

This is an example of a broader principle often summarised by **Goodhart's Law**: *When a measure becomes a target, it ceases to be a good measure*. In AI, optimising the benchmark is not always the same as improving intelligence.

Typical example

A language model achieves higher benchmark scores, yet human users observe no improvement in real-world usefulness.

Audit question

Does improving the metric genuinely improve the underlying capability?

Summary of Layer 6

Layer 6 is a checklist for scientific judgement. Each failure mode represents a question that every critical reader should ask.

Failure Mode	The Central Question
Vacuous Bounds	Is the theorem informative in practice?

Hidden Assumption	Which assumptions are doing the real work?
Scope Creep	Does the paper claim more than it proves?
Notational Sleight-of-Hand	Is every mathematical object well defined?
Selective Formalisation	Has the hardest part actually been formalised?
Reference Inflation	Are citations supporting the proof or decorating it?
Cherry-Picked Baselines	Are the comparisons fair?
Definitional Drift	Are key concepts used consistently?
Confused Causality	Does the evidence justify causal conclusions?
Goalpost Movement	Did the hypothesis change after seeing the results?
Overstated Novelty	What is genuinely new?
Metric Gaming	Does the metric reflect the real objective?

Completing the Framework

Together, the six layers form a complete method for analysing mathematical AI papers:One way to think about these six layers is as moving from **understanding** to **judgement**. The first layers help you identify the object, frame and claims; the middle layers explain how those claims are established and how they relate to practice; the final layer develops the habit of critically evaluating the work. By the time you complete all six layers, you are no longer simply reading an AI paper—you are analysing it as a mathematician, a scientist and a reviewer would.

- **Layer 1 – Mathematical Object** - What mathematical object is actually being studied?
- **Layer 2 – Theoretical Frame** - Which mathematical tradition is being used to study it?
- **Layer 3 – Claim Structure** - What exactly is being claimed, under what assumptions, and how broadly?
- **Layer 4 – Proof Architecture** - What mathematical ideas and techniques make the proof work?
- **Layer 5 – Empirical–Theoretical Bridge** - How well does the mathematics explain the empirical system?
- **Layer 6 – Methodological Audit** - How much confidence should we place in the paper's conclusions?